OFFICIAL TEACHER GRADING RUBRIC

Assignment 3: Matrix Spectral Decomposition | Total Marks: 5.0

Mark Breakdown (Total 5 Marks):

1. **Characteristic Equation Setup (2.0 Marks):**
   The student must correctly state the characteristic matrix equation $\det(A - \lambda I) = 0$ with matrix $\begin{pmatrix} 4-\lambda & 1 \\ 2 & 3-\lambda \end{pmatrix} = 0$.

2. **Characteristic Polynomial Expansion (2.0 Marks):**
   The student must expand $(4-\lambda)(3-\lambda) - (1)(2) = 0$ and simplify to the quadratic polynomial $\lambda^2 - 7\lambda + 10 = 0$.

3. **Factorization & Eigenvalues (1.0 Mark):**
   The student must factorize $(\lambda - 5)(\lambda - 2) = 0$ and state the correct eigenvalues $\lambda_1 = 5, \lambda_2 = 2$.